ASSIGNMENT-1

Variance and Bias (Diagram, overfit, underfit)

-For best fit model should we have low bias or high variance, low bias or low variance, high bias or high variance, low bias or high variance

Introduction to Bias and Variance

1.1 Introduction

In Machine Learning, the main objective of any model is to make accurate predictions on unseen data. However, while training a model, errors occur. These errors mainly arise from two important sources:

- **Bias**
- **Variance**

Understanding these two concepts is essential because they directly affect:

- Model performance
- Overfitting
- Underfitting
- Generalization ability

The relationship between bias and variance is called the **Bias–Variance Tradeoff**.

1.2 What is Bias?

Bias refers to the error caused by overly simplifying the machine learning model.

Definition:

Bias is the difference between the average prediction of the model and the true value.

In simple words:

Bias measures how much the model assumptions prevent it from capturing the true relationship.

Characteristics of High Bias

- Model is too simple
- Makes strong assumptions about data
- Cannot capture complex patterns
- High training error
- High testing error

This condition leads to **Underfitting**.

1.3 Underfitting (High Bias Case)

Underfitting happens when the model:

- Fails to learn patterns from training data
- Performs poorly even on training dataset

Example:

Using a straight line (Linear Regression) to fit highly curved data.

The model cannot represent the actual pattern → High Bias.

2.Understanding Variance

2.1 What is Variance?

Variance refers to how much the model's prediction changes when trained on different datasets.

Definition:

Variance measures the sensitivity of a model to small changes in the training data.

In simple words:

Variance tells us how much the model depends on the training data.

Characteristics of High Variance

- Model is too complex
- Captures noise in data
- Very low training error
- High testing error

This condition leads to **Overfitting**.

2.2 Overfitting (High Variance Case)

Overfitting happens when:

- Model memorizes training data
- Learns noise instead of pattern
- Performs well on training data
- Performs poorly on new data

Example:

Using a 10th-degree polynomial for only 10 data points.

The model passes through every point exactly → High Variance.

3.Bias–Variance Tradeoff with Diagram Explanation

3.1 Error vs Model Complexity Diagram

When we increase model complexity:

- Bias decreases
- Variance increases

The graph typically looks like:

- Left side → High Bias (Underfitting)
- Right side → High Variance (Overfitting)
- Middle → Best Fit

At the optimal point:

- Test error is minimum
- Bias and Variance are balanced

3.2 Target Board Diagram Explanation

A common way to explain bias and variance is using a dartboard example:

1. High Bias, Low Variance

- All points far from center
- Model consistently wrong
- Underfitting

2. Low Bias, High Variance

- Points scattered around center
- Model inconsistent
- Overfitting

3. Low Bias, Low Variance

- Points tightly grouped at center
- Accurate and consistent
- Best model

3.3 Mathematical Representation

Total Error =

$$\text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

Where:

- Bias^2 → Error from wrong assumptions
- Variance → Error from data sensitivity
- Irreducible error → Noise in data

4.Comparison Between Overfitting and Underfitting

4.1 Real-Life Example

Student Performance Prediction

- If we use only 1 feature (study hours) → Model too simple → High Bias
- If we use 100 irrelevant features → Model too complex → High Variance

Best model uses relevant features and proper complexity.

4.2 How to Reduce Bias

- Increase model complexity
- Add more relevant features
- Reduce regularization

4.3 How to Reduce Variance

- Increase training data
- Use regularization (L1, L2)
- Pruning in decision trees
- Cross-validation
- Dropout in neural networks

5.Best Fit Model and Final Conclusion

5.1 For Best Fit Model, What Should We Have?

Let us analyze the options:

1. Low Bias or High Variance ❌ (Overfitting)
2. High Bias or Low Variance ❌ (Underfitting)
3. High Bias or High Variance ❌ (Worst case)
4. Low Bias or Low Variance ✅ (Correct)

✅ **Correct Answer:**

For the best fit model, we should have **Low Bias and Low Variance**.

5.2 Why Low Bias and Low Variance?

- Low Bias → Model captures true pattern
- Low Variance → Model generalizes well
- Balanced complexity
- Minimum test error

This ensures:

- Good training performance
- Good testing performance
- Better generalization

5.3 Importance in AI and ML

Bias and variance concepts are important in:

- Regression models
- Classification models
- Neural networks
- Decision trees
- Ensemble learning

They help in:

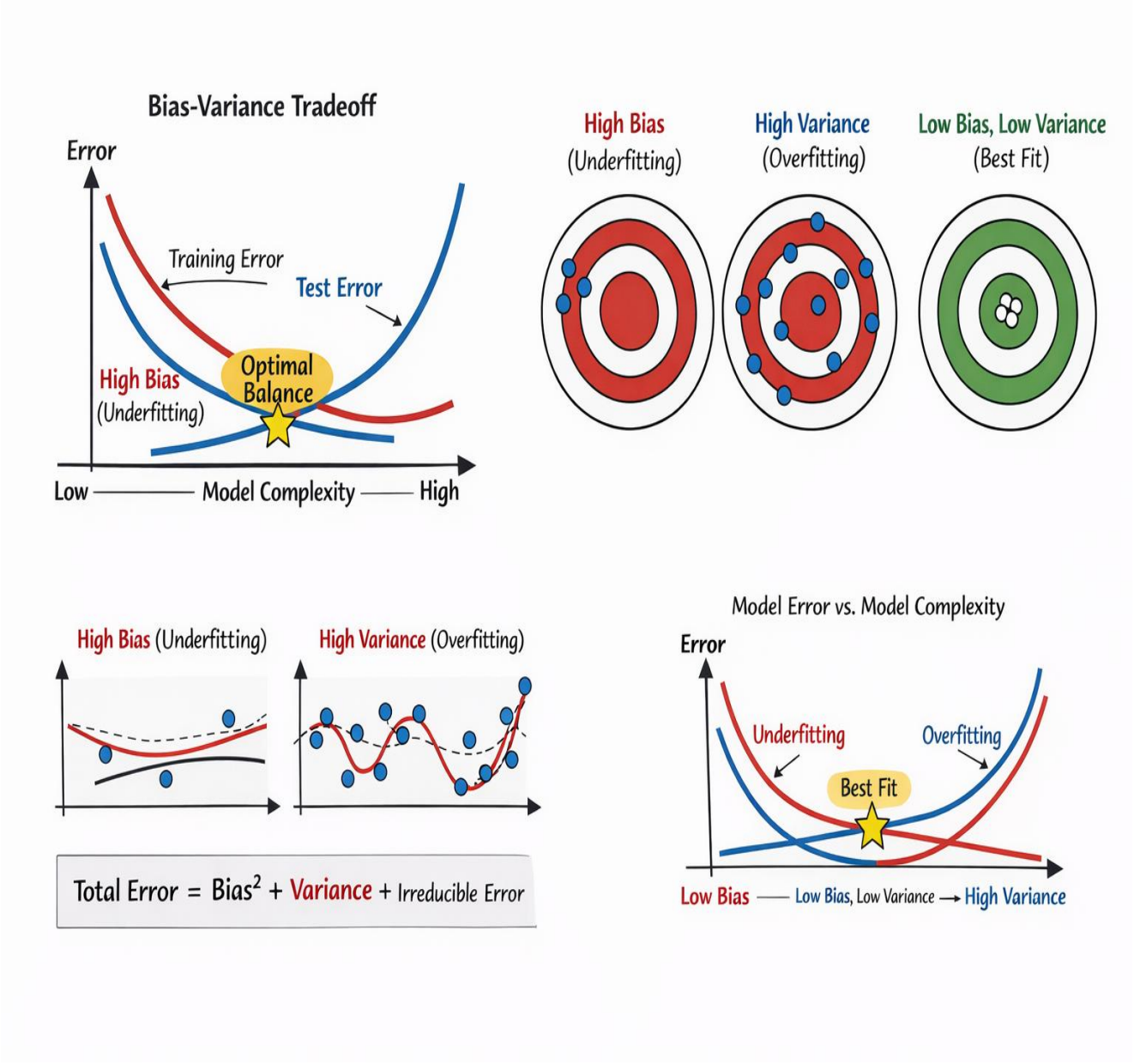
- Model selection
- Hyperparameter tuning
- Improving accuracy

- Avoiding overfitting and underfitting

Final Conclusion

Bias and Variance are two major sources of error in Machine Learning.

- **High Bias → Underfitting**
- **High Variance → Overfitting**
- **Low Bias + Low Variance → Best Fit Model**

The goal of every machine learning model is to find a balance between bias and variance, which is called the **Bias–Variance Tradeoff**.

Achieving this balance ensures that the model performs accurately on both training and unseen data.